MILITARY SPECIFICATION

FUZE, POINT DETONATING, PRACTICE, M751 WITH PYROTECHNIC SIGNATURE CARTRIDGE

This Amendment forms a part of Military Specification MIL-F-48637 (AR), dated 2 July 1987, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Paragraph 3.6, delete in its entirety and substitute:

"3.6 Proving ground test. Loaded fuzes from the Transportation Vibration Test, when ballistically fired, shall provide a visible signature at a "naked eye" viewing range of 1500 meters. An audible signature only on impact is not sufficient to meet this requirement."

PAGE 4

Add paragraph 3.13 Statistical Process Control (SPC). as follows.

"3.13 Statistical Process Control (SPC). This specification requires the use of SPC (see 4.1.2.2)."

Add paragraph 3.14 Kitting. as follows:

"3.14 Kitting. This specification requires the use of kitting (see 4.1.23)."

Paragraph 4.1.2 shall now end after the title. Prior to the word "Alternative" add 4.1.2.1 "Standard Procedure."

AMSC N/A FSC 1390

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Add paragraph 4.1.2.2 Use of SPC. as follows:

"4.1.2.2 Use of SPC. While it is desirable (optional) to use SPC as an inspection procedure on all parts, components, sub-assemblies, assemblies and complete rounds, it is mandatory that it be used on the following parts as indicated in their individual classification of defects paragraphs:

a.	Pin, setback	dwg. 9340437
b.	Slider	dwg. 9343999
c.	Striker	dwg. 9343995
d.	Spring, setback pin	dwg. 9343992"

Add paragraph 4.1.2.3 Use of kitting. as follows:

"4.1.2.3 Use of kitting. As with SPC, it is desirable (optional) to use kitting on all parts, it is mandatory to use kitting on the following parts to assure that they are not missing (their presence is the fuze is accounted for):

a.	Slider	dwg. 9343999
b.	Spring, slider	dwg. 9343994
c.	Rivet	dwg. MS9403
d.	Pin, setback	dwg. 9390437
e.	Spring, setback pin	dwg. 9343992"

PAGE 11

Table I, for Transportation Vibration Test, under PARAGRAPH REFERENCE/INSPECTION METHOD, delete "Test/4.5.5" and substitute "Test/4.5.6".

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Paragraph 4.4.1 d., delete: "twenty thousand fuzes" and substitute "twenty-six thousand fuzes."

Paragraph 4.4.2, delete in its entirety and substitute the following:

"4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, inspection plans shall be in accordance with the following (as applicable):

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a. For standard inspection procedures major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. All sampling plans are accept on zero defects and reject on one defect unless noted.

b. For examinations under SPC, the following shall be used:

"1. Quality conformance inspection utilizing SPC. The contractor may choose to utilize SPC on any or all parts; however, its use is mandatory on those parts specified in 4.1.2.2. Prior to the first article inspections and tests, the contractor shall submit an SPC implementation plan for approval by the government. The plan shall provide as a minimum, information regarding the following:

(a) Procedures including organizational/personnel responsibilities to be followed to demonstrate that the process is controlled.

(b) Procedures including organizational/personnel responsibilities to be followed after process is in control.

(c) Procedures including organizational/personnel responsibilities to be followed if something happens and the process goes out of control and what must be done to demonstrate that the problem has been corrected and the process is back in control."

"2 Acceptance criteria under SPC. Parts that are being manufactured under SPC do not have to undergo acceptance inspection in accordance with a specific sampling plan. Under SPC, acceptance shall be based on the following criteria:

(a) The process is in control (all points on the control chart are within the control limits).

(b) Both C_p and C_{pk} are equal to or greater than 1.33 where:

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C_p = $(USL - LSL)/(6\bar{R}/d_2)$

C_{pk} = the samller of the two values C_{pu} or C_{pl}

C_{pu} = $(USL - \bar{\bar{X}})/3\bar{R}/d_2)$

C_{pl} = $(\bar{\bar{X}} - LSL)/3\bar{R}/d_2)$

USL = Upper Specification Limit

LSL = Lower Specification Limit

R = Range = $X_{max} - X_{min}$ of the group of n being inspected

$\bar{R}$ = The Average Range of k groups that were inspected

$(R_1 + R_2 + \ldots R_k)/k$

$\bar{\bar{X}}$ = The Grand Mean of k groups that were inspected =

$(X_1 + X_2 + X_k)/k$

where $n_1 = n_2 \ldots n_k$

X = The individual measurements

$\bar{X}$ = The mean of the individual measurements

d = A common conversion factor which can be found in any statistics text. Common values are shown below on Table I

TABLE I

Sample Size	Factor
2	1.128
3	1.693
4	2.059
5	2.326
6	2.534

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It should be noted that until the SPC acceptance criteria above have been met by the process, acceptance of materiel will be in accordance with 4.4.2 a. above. Further, when the process goes out of control (i.e., one or more points are outside of the control limits or 1 or more points are outside of the specification limits), acceptance will go back to the requirements of 4.4.2 a. until the contractor has demonstrated that the process is back in control and the requirements for SPC as stated in this paragraph above have been met. Any point outside of the specification limits is cause for rejection of all material since the last inspection that was found acceptable for that particular examination."

PAGE 13

Paragraph 4.4.2.1, under notes, add the following:

"a. At the contractor's option, all "as cast" body features and/or dimensions that are not subject to any machining, shall be examined for their particular feature and/or dimension as defined in the paragraph, at the rate of 10 sample bodies randomly pulled from each die cavity per body lot. Any defects found in any of the 10 each die cavity samples shall be cause for rejection of all bodies produced from the particular die cavity.

b. All body features and/or dimensions that are machined, shall be examined per the criteria found in the appropriate classification of defects listed."

PAGE 15

Add to Minor 204, under NO. OF SAMPLE UNITS: "(a)."

PAGE 18

Major 101 under Requirement, delete "3.2" and substitute "3.12."

Delete from Majors 101, 103, 104, under NO. OF SAMPLE UNITS: "(a)."

Add to Majors 101, 103, 104, under AQL: "0.40%."

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Delete from Minor 201, under NO. OF SAMPLE UNITS: "(a)."

Add to Minor 201, under AQL: "0.65%."

PAGE 19

Delete from Majors 101, 102, 103, under NO. OF SAMPLE UNITS: "(a)."

Add to Majors 101, 102, 103, under AQL: "0.40%."

Delete from Minor 201, under NO. OF SAMPLE UNITS: "(a)"
Add to Minor 201, under AQL: "0.65%."

PAGE 20

Paragraph 4.4.2.7, delete Major 104 in its entirety.

PAGES 21-24

Paragraph 4.4.2.8, Majors 102, 103, under AQL, delete: "0.40%" and substitute: "b."

Paragraph 4.4.2.8, under NOTES, add "(b) SPC is required. See paragraph 3.13 and 4.4.2 b."

Paragraphs 4.4.2.9, 4.4.2.10, 4.4.2.11, except for Major 105, in NO. OF SAMPLE UNITS block, add: "(b)."

Paragraphs 4.4.2.9, 4.4.2.10, 4.4.2.11; under NOTES add: "(b) For spring sample size use MIL-STD-105D, Special Inspection Level 'S-3'."

PAGE 22-26

Paragraphs 4.4.2.9, 4.4.2.10, 4.4.2.11, 4.4.2.12, 4.4.2.13, under CATEGORY: Change "Monor" to "Minor."

PAGE 25

Paragraph 4.4.2.12, change Major 102 to Minor 202 and change AQL to read "0.65%."

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Paragraph 4.4.2.16, under CATEGORY: Change "Monor" to "Minor."

Paragraph 4.4.2.16, Major 101, AQL delete: "0.40%" and substitute: "a."

Paragraph 4.4.2.16, under NOTES add: "(a) SPC is required. See paragraph 3.13 and 4.4.2 b."

PAGE 30

Paragraph 4.4.2.17, Major 101, under AQL delete: "0.40%" and substitute: "a."

Paragraph 4.4.2.17, under NOTES add: "(a) SPC is required. See paragraph 3.13 and 4.4.2 b."

PAGE 31

Paragraph 4.4.2.18, SPECIAL A, under AQL delete: "0.40%" and substitute: "a."

Paragraph 4.4.2.18, Major 102, under AQL delete: "0.40%" and substitute: "a."

Paragraph 4.4.2.18, Major 104, under EXAMINATION OR TEST, delete entry in its entirety and substitute the following:

"Length between aft end and stop shoulder"

Under AQL delete: "0.40%" and substitute: "a"

Paragraph 4.4.2.18, under NOTES add: "(a) SPC is required. See paragraph 3.13 and 4.4.2 b."

PAGE 24

Paragraph 4.4.2.20, under Critical 1, delete entry in EXAMINATION OR TEST and substitute the following: "Length from aft end of striker to forward wall of signature charge cavity"; under PARAGRAPH REFERENCE/INSPECTION METHOD add after "Gage": "(b)."

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Paragraph 4.4.2.20, add Critical: "4"; under EXAMINATION OR TEST: "Height of striker over max from flange"; under AQL OR 100%: "100%"; under REQUIREMENT PARAGRAPH: "3.2"; under PARAGRAPH REFERENCE/INSPECTION MEHTOD: "Gage."

Paragraph 4.4.2.20, add under NOTES: "(b) The gage must be capable of making all acceptable fuzes before the fuze is removed from the gage system. Rejected fuzes shall not be marked and shall be removed from the lot and disassembled or destroyed at the contractor's discretion."

PAGE 35

Paragraph 4.4.2.21, delete Critical 1 in its entirety and substitute the following under CATEGORY: "Special A"; under EXAMINATION OR TEST: "Red striker band is visable on fuze"; under AQL OR 100%: "100%"; under REQUIREMENT PARAGRAPH: "3.2"; under PARAGRAPH REFERENCE/INSPECTION METHOD: Visual."

PAGE 36

Paragraph 4.4.2.22, delete Critical 1 in its entirety and substitute the following: under CATEGORY: "Special A"; under EXAMINATION OR TEST: "Red striker band is visable on fuze"; under AQL OR 100%: "100%; under REQUIREMENT PARAGRAPH: "3.2"; under PARAGRAPH REFERENCE/INSPECTION MEHTOD: "Visual."

PAGE 37

Last paragraph on page, delete "4.5.1.2.2 Damaged fuze" and substitute "4.5.1.2.3 Damaged fuze."

PAGE 38

Paragraph 4.5.2, add the following at the end of the paragraph: "This is a destructive test. Primers subjected to this test shall not be returned to the lot."

Paragraph 4.5.3, add the following at the end of the paragraph: "This is a destructive test. Springs subjected to this test shall not be returned to the lot."

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Paragraph 4.5.4, add the following at the end of the paragraph: "This is a destructive test. Fuzes subjected to this test shall not be returned to the lot."

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Paragraph 4.5.5 b., add the following at the end of the sentence: "Fuzes subjected to this test shall not be returned to the lot."

Paragraph 4.5.7, delete "or M29" from the third line of the second paragraph.

Paragraph 4.5.7, under firing schedule, delete subparagraph "b" in its entirety and substitute the following:

"b. Twenty (20) rounds at charge "1" at a gun elevation of forty-five (45) degrees."

PAGE 40

Paragraph 4.5.7, under acceptance/rejection criteria, delete all criteria in its entirety and substitute the following:

"a. All charge "0" firings shall be for information only.

b. For the charge 1 and charge 4 firings, two or more duds (total duds for both groups combined) constitutes cause for lot rejection.

A dud is defined as an unobservable signature upon ground impact of the round. Observable is defined as both visual and audible."

PAGE 41

Paragraph 4.5.8, add the following after the last sentence:

"Fuzes subjected to this test shall not be returned to the lot."

Custodian:
Army-AR

Preparing activity:
Army-AR

Project 1390-A687)

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